

Apollo 11 spacecraft at top weighs 97,000 lb (44,000 kg)

SATURN V

The most powerful rocket ever built, the Saturn V, was masterminded by the German scientist Werner von Braun in the United States and was used to launch the Apollo astronauts on eight missions to the moon between 1967 and 1972. Later versions of the rocket were used to launch Skylab—the first American space station—and the American part of the Apollo-Soyuz joint mission.

Third stage, with single engine burning liquid hydrogen and liquid oxygen

Second stage, with five engines burning liquid hydrogen with liquid oxygen

Tower holds rocket upright before launch

First stage, with five engines burning kerosene with liquid oxygen

Total launch vehicle with three separate rocket stages is 363 ft (110.6 m) high

▼ FALCON 9 LANDING

The first stage of a SpaceX rocket makes a vertical touchdown after completing its role in a satellite launch.



REUSABLE ROCKETS

The enormous cost of rocket stages destroyed or lost during launch is a major barrier to space exploration. NASA's Space Shuttle (see p.273) had some reusable elements. Since 2016, the SpaceX company has made huge advances in recovering the stages of its Falcon rocket for future missions.

MULTISTAGE ROCKET

Firing multiple rockets in sequence is the most efficient way to get heavy loads into space. As each rocket stage uses up its fuel supply, it detaches and falls back to Earth, reducing the amount of empty weight the rest of the rocket has to push forward. The cargo that eventually reaches orbit is known as the rocket's payload.

